

STMBAD: Spatio-Temporal Multimodal Behavior Anomaly Detector for Industrial Control Systems

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Abstract—Modern cyber attacks against industrial control systems (ICSs) are highly stealthy, persistent, and targeted. Existing anomaly detection methods are mainly based on a set of rules defining correct behaviors and use loosely bounded detection thresholds, which can be exploited by attackers to evade detection. In this article, we propose STMBAD, a spatio-temporal multimodal behavior anomaly detector based on spatio-temporal ICS behavior analysis to improve the performance of ICS anomaly detection. STMBAD leverages the rich information available in industrial multimodal data to achieve a deep understanding of complex ICS behaviors and enhance the ability to detect stealthy attacks. To avoid data processing cross heterogeneous type/structure and temporal confusion caused by unsynchronized time series, STMBAD embeds time series of individual modality separately into variate tokens and applies the attention mechanism and feedforward network to capture multivariate correlations and interdependencies. Meanwhile, based on the attention mechanisms, temporal evolution law and spatial correlation of different modalities can be captured to model the characteristics of the spatio-temporal multimodal behavior of ICS. When detecting attacks, an adaptive detection mechanism combining global and local detection is proposed to utilize dynamic thresholds at different levels and reduce errors caused by a loose global threshold. The simulation results show that the proposed method outperforms the baseline methods and yields the highest F1 score, reaching 95%.

Index Terms—Adaptive thresholds, anomaly detection, industrial control system (ICS), multimodal data.

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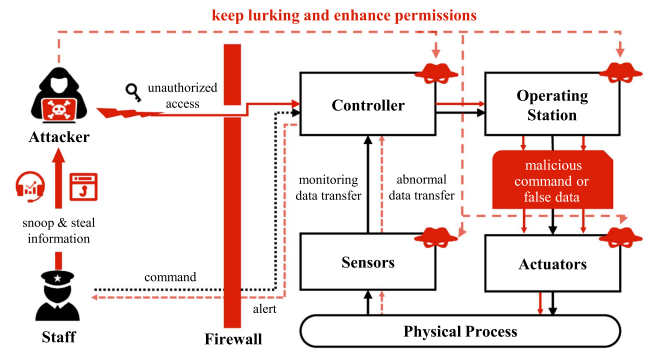


Fig. 1. Attacks in ICS environments.

I. INTRODUCTION

INDUSTRIAL Internet is a promising technology that combines industrial systems with Internet connectivity to improve production efficiency and reduce production costs. Industrial control systems (ICSs) are important in the industrial Internet, providing process automation, distributed control, and process monitoring. To fulfill the requirements of the industrial Internet, ICSs are connected to enterprise networks, allowing business users to access real-time data generated by industrial systems. However, this new paradigm exposes insecure devices to public networks and raises security issues as those devices are more vulnerable to external attacks [1].

Anomaly detection is widely used to identify potentially anomalous activities of a system when a lurking adversary or malware starts to launch an attack on the system. Existing anomaly detection technology solutions can be categorized into mechanism-driven and data-driven methods. Mechanism-driven approaches leverage expert knowledge of the underlying mechanisms to detect anomalies [2]. These approaches require comprehensive theoretical knowledge that is difficult to obtain for highly complex ICSs. Data-driven anomaly detection methods use techniques such as statistical analysis, data mining, and machine learning to automatically identify anomalous data points or events that are significantly different from normal patterns from a large amount of collected data [3]. The data-driven methods are widely used because they do not need to consider the increasingly complex physicochemical reaction mechanisms in industrial processes.

However, since modern attacks against ICSs are usually stealthy, persistent, and targeted, it is much difficult to detect them. As shown in Fig. 1, attacker lurks in the target system for a long time and uses a combination of attacks to gain control or

modify data by exploiting vulnerabilities in the system, such as in the SolarWinds attack [4] and the Stuxnet attack [5]. Sophisticated attackers usually have a deep understanding of the execution of the targeted ICSs and can hide their manipulation in the system by closely following the expected behavior of the system while injecting just enough false information at each time step to achieve their goals [6]. Previous methods leverage a set of rules defining the correct behavior of ICSs. However, the rules define variable ranges across all possible ICS states, leading to loosely bounded detection thresholds. Thus, attackers can exploit these loose bounds and launch stealthy attacks that evade detection without violating such rules.

To address these problems, anomaly detection systems need to understand not just individual actions but also complex sequences of events over time so as to detect subtle changes in behavior that might be indicators of compromise. Thus, this work leverages multimodal data to identify spatio-temporal behavior patterns that might indicate sophisticated attacks in ICSs. Industrial multimodal data are collected from multiple sources and commonly exhibit a large degree of heterogeneity in form, scale, and resolution [7]. The fusion of multimodal data can reveal the operational characteristics of the system more accurately, hence effectively improving the accuracy of data analysis, such as quality prediction [8] and anomaly detection [9]. Instead of just looking at instantaneous values, we consider the trajectory of variables over time, which can help detect subtle deviations that might indicate a stealthy attack. Meanwhile, we also consider correlations between different components or sensors in the ICS, which can help detect attacks that manipulate multiple variables in concert to stay within individual thresholds.

Based on a better understanding of ICS behaviors, we develop algorithms to discover recurring patterns in the spatio-temporal behavior of the system and monitor deviations from these patterns to indicate potential attacks. Leveraging the rich information available in ICS environments can significantly enhance our ability to detect and respond to complex, stealthy attacks. However, developing an anomaly detector for ICS that monitors spatio-temporal multimodal behavior is challenging, at least from the following perspectives.

- 1) *Multimodal data are heterogeneous*: The multimodal data are heterogeneous in data type and structure, which poses huge challenges to the development of data-driven models. Effective solutions have been proposed to address the issues arising from multimodal data processing, such as missing values resulting from heterogeneous sampling sequences [10]. However, one important problem in multimodal data preprocessing is to integrate diverse data types (continuous, discrete, categorical) from multiple sources with different formats and structures. The state-of-the-art multimodal data analysis models, such as transformer-based approaches [11] embed multiple variates of the same timestamp into indistinguishable channels and apply attention to these temporal tokens to capture temporal dependencies. The unified embedding for each temporal token that fuses multiple variates with potentially unaligned timestamps may fail in learning variate-centric representations and result in meaningless attention maps.
- 2) *Multimodal data are unsynchronized*: Temporal alignment and synchronization are critical challenges in

multimodal learning, especially when dealing with data from ICS where various sensors and subsystems may operate at different sampling rates and with different timing constraints. When multimodal data are unsynchronized, the data points from a single time step cannot reveal beneficial information due to the temporal confusion between multivariate. Besides, while modalities can be greatly influenced by the sequence order, attention mechanisms are improperly adopted on the spatial dimension to analyze the multivariate correlations. These issues can significantly impact the accuracy and reliability of data analysis and decision-making processes.

- 3) *Threshold setting is critical to the effectiveness of anomaly detection*: The spatio-temporal multimodal behavior changes with the manufacturing process. Setting appropriate thresholds for anomaly detection is often a balancing act. Too sensitive thresholds lead to false alarms, while too lax thresholds result in missed detections. Most existing anomaly detection methods use static thresholds to determine anomaly results. However, the operating conditions, data distribution, and environmental factors of ICSs would change according to manufacturing processes. Thus, static thresholds cannot adapt to these dynamic changes and lead to false alarms or missed detections [12]. Furthermore, due to ignoring spatio-temporal dependencies, static thresholds cannot capture contextual anomalies [13].

To address these challenges, this article proposes a spatio-temporal multimodal behavior anomaly detector (STMBAD) for ICSs based on the spatio-temporal behavior analysis of multimodal sequences. For challenges (1) and (2), STMBAD employs inverted transformers (iTransformer) framework [14] to industrial multimodal data. In the iTransformer, the roles of the attention mechanism and the feedforward network (FFN) are inverted. The time points of individual series are embedded into *variate tokens*, which are then utilized by the attention mechanism to capture multivariate correlations. Meanwhile, the FFN is applied to each variate token to learn nonlinear representations. STMBAD embeds individual series into variate tokens and uses the attention mechanism to capture multivariate correlations, thereby preserving the internal feature information of the modality and avoiding the problem of temporal misalignment. For challenge (3), which is about the threshold setting to enable effectiveness of ICS anomaly detection, STMBAD obtains the sliding window-based loss values by comparing the reconstructed data within the sliding window with the real data. To better adapt the detection method to the changing data distribution, the model updates the loss value of each window of the test data to the loss value sequence of the training data in real time. Then, the percentage method is used to determine the threshold value of the current window, thus realizing adaptive anomaly detection.

The main contributions of this work are as follows.

- 1) *A method for analyzing the spatio-temporal correlation of industrial multimodal data is proposed*: The proposed method embeds the time points of individual series into variate tokens and uses the attention mechanism to capture multivariate correlations, weakening the impact of temporal interference and confusion caused by modality heterogeneity and misalignment.

- 2) *An adaptive threshold mechanism combining global and local detection is proposed:* It ensures that the system can utilize dynamic thresholds at different levels and reduce errors caused by insufficient local information or by only processing global information.
- 3) *A multimodal data-based framework is developed to identify anomalous events in the industrial process by mining the spatio-temporal correlations between multivariate in production equipment.*

The rest of this article is organized as follows. Section II overviews the related work on ICS anomaly detection. Section III describes in detail the algorithm design of STMBAD. The evaluation is discussed in Section IV. Finally, Section V concludes this article.

II. RELATED WORK

The growing complexity of ICSs leads to the industrial data exhibiting multimodal, nonlinear, and strongly coupled characteristics [15], which poses new challenges to anomaly detection methods. In face of these challenges, statistical learning, machine learning, deep learning, and transformer-based methods have emerged as widely employed approaches of anomaly detection based on multimodal data.

Statistical learning methods: Traditional techniques such as cluster-based methods [16], independent component analysis (ICA), and rule-based approaches [2] have been applied to anomaly detection in multimodal data. Minjie et al. [17] extracted the key features of traffic through principal component analysis and constructed a normal traffic behavior detection model for power IoT terminals based on one-class support vector machines (OCSVMs). Harrou et al. [18] combined ICA and Kantorovich distance-based processing of supervisory control and data acquisition data with kernel density estimation to compute thresholds. This makes the model more flexible in detecting sensor anomalies such as accuracy degradation and freezing faults. However, the thresholds of this approach are determined based on residuals during the training phase and cannot be dynamically adjusted to adapt to the changing state of the system. Moreover, these methods also suffer from information loss when dealing with complex data distributions and high-dimensional data [19], which significantly degrades the performance of ICS anomaly detection.

Machine learning methods: Machine learning algorithms can automatically learn the process dynamics and control strategies deployed in an ICS, which leads to relatively easier and faster creation of anomaly detectors [20]. Li et al. [21] proposed a parallel mutual information computation based on the Spark platform, which resulted in a weighted outlier attribute detection scheme for mixed data and reduced the detection time of outliers. In response to the fact that OCSVM is not suitable for feature extraction for high-dimensional data, He et al. [22] proposed combining dictionary learning with OCSVM to extract correlations and local dynamics in multivariate time series for sparse representation and anomaly detection tasks. The accuracy of these models is highly dependent on the quality of the input data. Moreover, either undersampling or oversampling of multimodal data affects model performance [23]. Xu et al. [24] addressed this problem by using a synthetic few oversampling technique algorithm and random undersampling to balance the dataset. They also reduced the model by discarding features with low

computational mutual information to reduce computational cost and improve overall performance.

Deep learning methods: Recently, deep learning has been widely used due to its powerful feature extraction and representation learning capabilities [25]. For example, Zheng et al. [26] integrated correlation learning into a spatio-temporal network of multivariate time series and uses graph convolutional networks to obtain spatial dependencies between variables and reconstruct data. Notably, this model can identify the root causes of anomalous events by ranking the univariate variables that contribute the most to the anomaly score, enabling early detection and interpretability of anomalies. However, it has limitations in capturing temporal evolution due to misalignment of multimodal time series and distortion of internal modal features. Therefore, Padmavathi et al. [27] combined the attention mechanism with bidirectional LSTM and uses CNN to extract pattern features to improve the accuracy of malicious node detection. In [28], an efficient real-time sensor anomaly detection method is used to detect instantaneous network anomaly information by recognizing sensor anomalies through Markov and LSTM-based networks.

Transformer-based methods: However, in real industrial scenarios, the data collected by sensors can have information loss due to a series of complex reasons. Meanwhile, real multidimensional time-series data have repetitive patterns in the long and short term, which is intractable for LSTM-based models. Therefore, many articles have used transformer to capture long-term dependencies in time series. To mitigate the overfitting problem, Miao et al. [29] used an autoencoder based on a transformer architecture to capture temporal dependencies in time series and integrates the encoder into a generative adversarial network to constrain the reconstruction process. Zhang et al. [30] introduced the Conv-Transformer model by combining local and global features of the image to achieve higher accuracy and use fewer computational resources. Ma et al. [31] learned the temporal patterns of transformer-based sequences and detects anomalies through pattern differences to achieve adaptive detection of the model. Although these models make up for the shortcomings of LSTM, they still force multiple variables under the same timestamp to be aligned into one token, which affects correlation extraction among variables to some extent and introduces unpredictable noise at the same time.

III. ALGORITHM DESIGN

A. Architecture Overview

The framework of STMBAD is composed of two main parts: *behavior modeling module and anomaly detection module*, as depicted in Fig. 2. Industrial multimodal data from multiple sources (referred to as modalities), which generally exhibit a significant degree of heterogeneity in form, scale, resolution, and accuracy [7], is input into STMBAD. The behavior modeling module is developed based on iTransformer framework [14]. In this module, the spatial-temporal correlation between data sequences is extracted to establish a spatio-temporal sequence behavior model of the normal industrial process. In anomaly detection module, a data reconstruction based anomaly detection module is implemented to monitor the spatio-temporal behavior of multimodal data. In this module, the test data is classified as benign or abnormal by using adaptive thresholds.

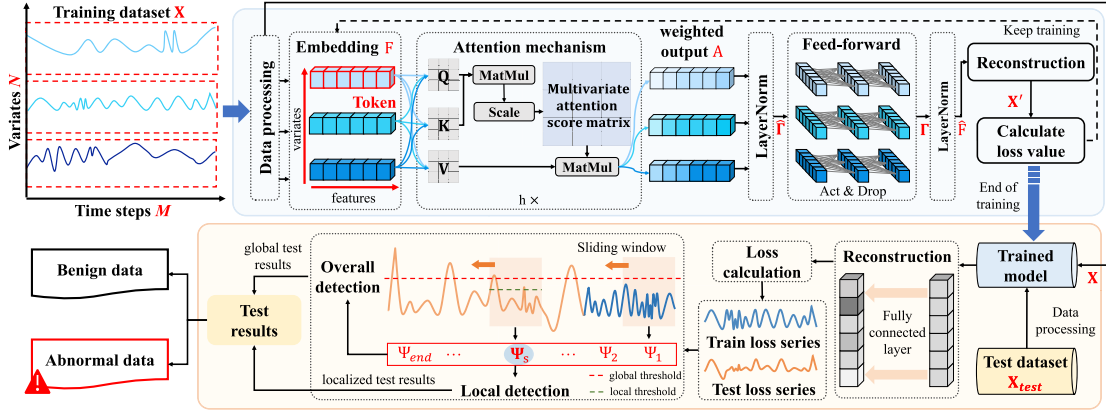


Fig. 2. Algorithm architecture overview. The blue area is the behavioral modeling part and the yellow area is the anomaly detection part. In the figure, $h \times$ represents an attention mechanism with a total of h attention heads. Q , K , and V denote Query, Key, and Value, respectively. Γ is the result of a nonlinear transformation of each input feature and X' is the result after reconstruction. Anomaly detection is a combination of global and local anomalies, where ψ denotes a local threshold within each window.

The training data is defined as $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\}$, where N is the number of modalities. The n th modality is $\mathbf{x}_n = \{x_n^1, x_n^2, \dots, x_n^M\}^T$ ($n \in [1, N]$) with M as the length of $\mathbf{x}_n$. To avoid information loss due to dimensionality reduction in data preprocessing, STMBAD directly performs an independent embedding operation on each sequence $\mathbf{x}_n$ by using a shared multilayer perceptron (MLP), denoted as $\mathbf{F} \in \mathbb{R}^{N \times D}$, and D is the embedding dimension. The test dataset can be denoted as $\mathbf{X}_{\text{test}} \in \mathbb{R}^{N \times M_{\text{test}}}$, which performs the same embedding operation as $\mathbf{X}$, and M_{test} is the length of test sequences. The behavioral modeling uses only the training dataset to learn the normal behavior of system, while anomaly detection is performed on the test dataset. Anomaly detection result is represented as a vector $\mathbf{L}' = \{\mathbf{L}'_t\}$ ($t = 1, 2, \dots, M_{\text{test}}$) consisting of predicted labels $\mathbf{L}'_t \in \{0, 1\}$ representing ICS is normal ($\mathbf{L}'_t = 0$) or abnormal ($\mathbf{L}'_t = 1$) at time t .

B. Behavioral Modeling

To capture the relationships between modalities, STMBAD employs a self-attention mechanism to compute the correlation among them, and utilizes a multihead attention mechanism to weight the values.

The queries, keys, and values $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N \times d_k}$ are obtained using linear projection, where d_k is the projected dimension. The i th attention head gets a set of $\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i$. The attentional weights are obtained by calculating the similarity between the query and the keys, using the attentional weight matrix to perform a weighted summation on the value matrix $\mathbf{V}$. The weighted output $\mathbf{A}$ is defined as

$$\mathbf{A} = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) \mathcal{M}$$

$$\text{head}_i = \mathbf{V}_i \left[\text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_i^T}{\sqrt{d_k}} + \text{mask} \right) \right], i \in [1, h] \quad (1)$$

where h denotes the number of attention heads and $\mathcal{M}$ is a learnable output weight matrix.

STMBAD treats the entire sequence of a modality as a token, as shown in Fig. 3, while the correlation analysis focuses on the interaction among modalities. Consequently, the $\mathbf{A}$ acquired

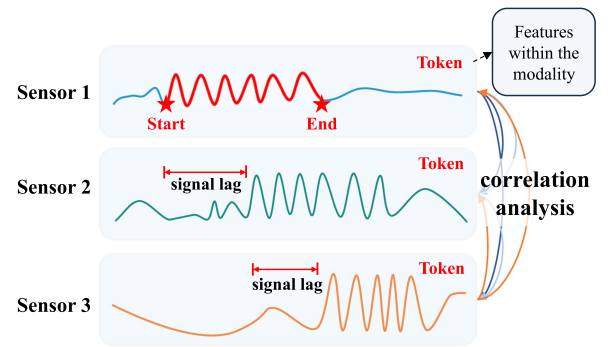


Fig. 3. STMBAD treats the entire sequence of a modality as a token, focuses only on the internal features of the modality during embedding, and then analyzes the correlation between modalities.

through the attention mechanism is a weighted representation integrating global information and the self-attention mechanism. This is because when Q and K perform the dot product operation, the information of one modality interacts with that of other modalities. The obtained attention score indicates, to a certain degree, the contribution of this variable to the entire system. In further interaction with the value V , variables with high correlation can obtain larger attention weights.

In the encoder, the self-attention mechanism is used to achieve correlation extraction between different variables and layer normalization is used to obtain the intermediate variable $\hat{\Gamma}$. Layer normalization is applied to the feature representation so that the feature channels of all variables are under a relatively uniform distribution. For each weighted token $\mathbf{A}_i$, perform Z-Score normalization as

$$\hat{\Gamma} = \frac{\mathbf{A}_i - \mu}{\sigma}, (i = 1, 2, \dots, N) \quad (2)$$

where μ and σ are the mean and standard deviation of $\mathbf{A}_i$.

Fully connected features are encoded for $\hat{\Gamma}$ in an FFN and the reconstructed result is obtained by projection and cropping. Different from the FFN in the traditional transformer, which encodes variables under the same time stamp, the FFN in this

Algorithm 1: Behavioral Modelling.

Data: training data ($\mathbf{X}$), train epoch (k), training window (ω) and tolerance number (Θ).
Result: Reconstructed data ($\mathbf{X}'$).

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1  $\mathbf{F} = \text{Embedding}(\mathbf{X})$ ;
   $\triangleright$  Embedding sequences into variate tokens
2  $\Theta = 3$ ;  $\triangleright$  early-stop tolerances triggered
3 for  $i \leq k$  do
4   foreach  $\omega$  do
5     Gets vectors  $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ ;
6      $\mathbf{A} \leftarrow$  Apply self-attention layer on variate
      tokens according to (1);
7      $\hat{\mathbf{F}} \leftarrow \text{LayerNorm}(\mathbf{A})$ ;  $\triangleright$  layer normalization
8     Use FFN to obtain sequence representations  $\mathbf{\Gamma}$ 
      according to (3);
9      $\hat{\mathbf{F}} \leftarrow \text{LayerNorm}(\mathbf{\Gamma})$ ;  $\triangleright$  Layer normalization.
10     $\mathbf{X}' \leftarrow \text{MLP}(\hat{\mathbf{F}})$ ;  $\triangleright$  according to (4)
11  end
12   $\text{Loss} \leftarrow \text{GetLoss}(|\mathbf{X}' - \mathbf{X}|)$ ;  $\triangleright$  according to (5)
13   $\triangleright$  Save Model;
14  if Meet the stopping condition then
15    if  $\Theta = 0$  then
16      Stop();  $\triangleright$  early termination
17    else
18       $\Theta \leftarrow \Theta - 1$ ;
19    end
20  end
21 end

```

article represents tokens for the entire sequence of variables and calculates the global representation within the sequence. As a part of each EncoderLayer, FFN consists of two fully connected layers each of which has an activation function. The input is linearly transformed by a one-dimensional convolutional layer, which converts the dimensionality of the input from D to a higher feedforward neural network dimensionality D' . This linear change can be expressed as $\mathbf{W}_1 \hat{\mathbf{F}} + \mathbf{b}_1$, where $\mathbf{W}_1$ is the weight matrix of convolutional layer and $\mathbf{b}_1$ is bias vector.

The dimension of the output is transformed from D' back to D by a second one-dimensional convolutional layer in order to add it to the Nonlinear rectified linear unit (ReLU) activation function is applied to the output of the convolutional layer. As a result, the entire computational process of the FFN can be represented as

$$\mathbf{\Gamma} = \text{ReLU}(\mathbf{W}_1 \hat{\mathbf{F}} + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2 \quad (3)$$

where $\mathbf{W}_2$ is the weight matrix of the second convolutional layer and $\mathbf{b}_2$ is the bias vector and get $\mathbf{\Gamma} \in \mathbb{R}^{N \times D}$.

After fully connected feature encoding using FFN, it is again normalized by one layer to obtain $\hat{\mathbf{F}}$. Finally, each variable representation is mapped to the reconstructed result separately through a mapping layer, which is implemented by MLP as

$$\mathbf{X}'_{N \times M} = \hat{\mathbf{F}}_{N \times D} \mathbf{W}_{D \times M} + \mathbf{b} \quad (4)$$

where $\mathbf{W}$ is the parameter that the model wants to learn and $\mathbf{b}$ is an M -dimensional bias vector. Projective mapping of the encoder's output yields the output $\mathbf{X}'$.

The behavioral modeling procedure for spatio-temporal multimodal time series is given in the Algorithm 1. The algorithm takes as input the training data $\mathbf{X}$, training epoch k , and training window size ω . The output is reconstructed data $\mathbf{X}'$. Algorithm 1 is an iterative process whose main process is multimodal time-series correlation analysis as lines 5–10. To avoiding overfitting, the process of training will be stopped when the model performance improves no longer. In each iteration, the loss function value is used as a performance indicator to compare the current validation loss with the previous minimum validation loss. There, we use the smooth L1 loss function to evaluate the difference between the model reconstructed values and the true values. The smooth L1 loss function is defined as

$$\text{Loss}(\varsigma) = \begin{cases} \frac{1}{2} \varsigma^2, & \text{if } |\varsigma| \leq 1 \\ |\varsigma| - \frac{1}{2}, & \text{otherwise} \end{cases} \quad (5)$$

where ς is the difference between the reconstructed value and the true value.

The best score is updated and the model weights are saved, as shown in lines 13–20. If the current validation loss is not significantly reduced or exceeds a certain tolerance range of the minimum validation loss, an early stopping condition is triggered. Meanwhile, a counter, denoted as Θ , indicating the triggering times of early stopping. The training will be stopped early when the counter exceeds the tolerance value.

The entire encoder can be represented as a stack of multiple encoder layers. In one encoder, by using MLP, nonlinear transformations and feature extraction can be performed on the inputs, resulting in a higher level representation of the input data. At the same time, global correlation analysis and encoding of features are performed in an inverted manner, avoiding information weakening and noise interference caused by temporal misalignment.

C. Industrial Process Anomaly Detection

Existing detection methods utilize the statistical characteristics of normal parameters to determine a fixed threshold value as the basis for system detection. However, when operating conditions or the environment change, the internal properties of the data will also change. The threshold value determined using historical data will be unable to capture such changes, leading to detection errors. Therefore, we implement a multiple detection strategy in the detection phase. This strategy combines historical and current information with a sliding window to ensure that the system can make full use of different levels of information. By doing so, it reduces errors that may be caused by insufficient local information or global information processed individually, thereby improving the comprehensiveness, accuracy, and robustness of the detection. In the real world, devices have a boot phase for software or firmware loading. Hence, we take 30% of the training set and use it to simulate the state changes during the boot phase.

Specifically, as shown in Fig. 4, multiple detection strategy includes global detection and local detection. For each time series $\mathbf{x}_n$, after reconstruction to obtain the loss sequence $\hat{\mathbf{x}}_n$, we employ the sliding window and the three-sigma rule for detecting the degree of deviation. Assume that the size of the sliding window is w and window slide step is m , where $m \leq w$. By sliding the window sequentially from the starting position of loss series. The subsequence can be recorded as $\hat{\mathbf{x}}_{[s:s+w-1]} =$

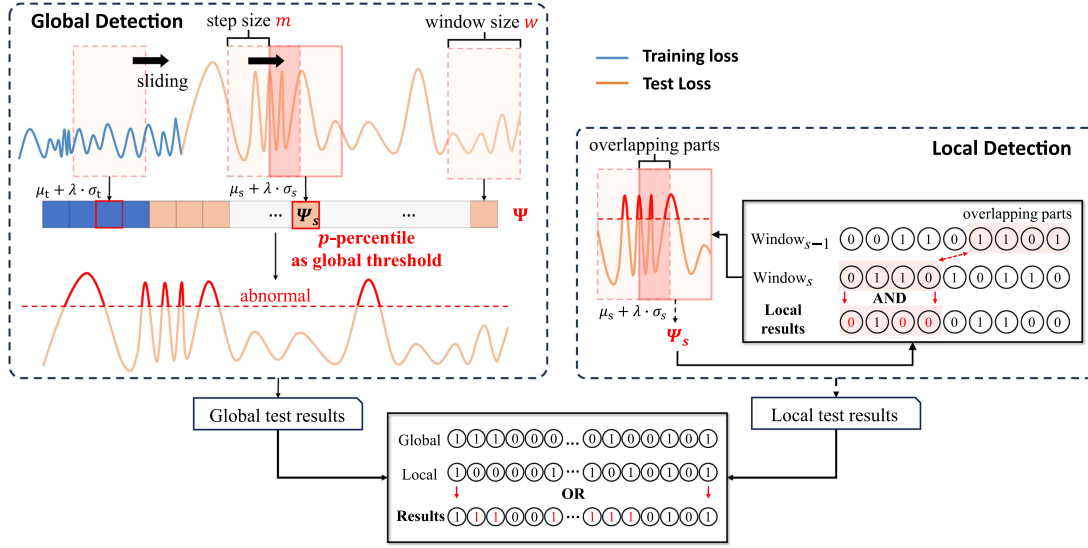


Fig. 4. Illustration of anomaly detection. In the figure, The window keeps sliding and each window outputs a local threshold value ψ_s . Then all these ψ_s values are connected together to obtain Ψ , where μ and σ denote the mean and standard deviation of the subsequence within the window, respectively. λ is the standard deviation multiplier, t and s denote the time point corresponding to the current window starting point, and w is the local threshold of the window $[s, s + w]$.

$\{\hat{x}_s, \hat{x}_{s+1}, \dots, \hat{x}_{s+w-1}\}$, the window mean μ_s and standard deviation σ_s are obtained. STMBAD measures the extent of deviation of the loss data within the window in terms of the three-sigma rule to obtain a measured value as

$$\psi_s = \mu_s + \lambda \sigma_s \quad (6)$$

where $\lambda = 3$. As a rule of thumb, most data points can be covered when $\lambda = 3$ to provide more conservative bounds [32]. Subsequently, STMBAD concatenates the obtained ψ_s to have a dynamic sequence of measurements Ψ .

For global detection, the percentage method is applied to determine the threshold. The percentage method is based on the principle of data distribution, where the dataset is sorted in ascending order and the p th percentile is selected as the threshold. If any point in $\hat{x}_n$ exceeds the threshold that point in time is marked as an abnormal. This method ensures that the threshold is set within a specific distribution of the data, thus effectively identifying extreme values or anomalies in the data.

In localized detection, the local threshold is ψ_s . Since $m \leq w$, this means that the window $\hat{x}_{[s:s+w]}$ partially overlaps with the previous window $\hat{x}_{[s-w-1:s-1]}$. The overlapping part undergoes repeated detection, and the results may show bias due to noise or sudden data changes. Therefore, to reduce the interference of uncertainty and enhance the robustness of the detection method, we apply the AND operation in the overlapping part to achieve stricter detection criteria. This implies that the overlap is considered an exception only when both results of repeated testing are abnormal.

Finally, in the face of the different results obtained from global and local detection, we use the OR operation. Local detection can provide fine-grained information, and global detection can provide macroscopic contextual information. The OR operation can combine local and global detection results. In this way, anomalies missed in local detection due to strict criteria can be supplemented by this operation. At the same time, possible misdetections in global detection can be corrected by local detection.

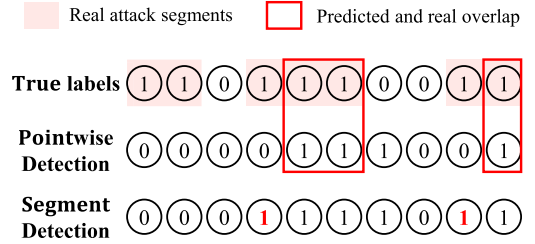


Fig. 5. Examples of segment detection. Adjustment of predicted labels when predicted anomalies overlap with real anomaly segments.

In practice, anomalies typically do not occur in isolation but rather manifest as a continuous segment of anomalies. Focusing solely on the detection of a single point may overlook the reality of the entire segment of anomalies. Therefore, STMBAD employs a segment detection strategy [33]. Fig. 5 shows that in this strategy, when any point in the actual abnormal segment is detected, the segment is considered to have been correctly detected. This approach reduces the reliance on isolated points and avoids having false alarms at a single point affect the overall assessment.

STMBAD proposes a multiple detection method with adaptive thresholds. This approach aims to enhance the accuracy and robustness of anomaly detection by capturing long-term trends and periodic changes in the system. It can flexibly and automatically adjust thresholds to adapt to different time scales and changing environmental contexts.

IV. EVALUATION

A. Datasets

The proposed STMBAD performed experimental simulations in industrial water treatment processes and water distribution (WADI) processes. Secure water treatment (SWaT), battle of attack detection algorithms (BATADAL), and WADI are used to evaluate the validity and performance. These scenarios simulate

TABLE I
CORE INFORMATION OF DATASETS

Dataset	Dimension	Training size	Test size	Anomaly rate
SWaT	51	496800	449919	12.140%
BATADAL	43	8761	4177	5.700%
WADI	123	784572	172801	5.774%

TABLE II
EXPERIMENTAL HYPERPARAMETERS

Dataset	Time step	Encoder layers	Batch size	Kernel size	Dimension	
					input	output
SWaT	100	3	128	6	128	128
BATADAL	100	3	1	6	64	64
WADI	100	3	64	6	128	128

physical attacks, such as water leaks and malicious data injection, in addition to attacks and defenses against programmable logic controllers (PLCs) and networks. The core information of these datasets is given in Table I.

The detailed description of each dataset is as follows.

- 1) *SWaT* [34]: SWaT is a water treatment testbed for research in the field of cybersecurity. During data collection, the ICS was run for 11 consecutive days. Of these, 7 days were in normal operation, and 4 days were in an attack scenario. Attacks last anywhere from a few minutes to an hour.
- 2) *WADI* [35]: WADI is a natural extension of SWaT, forming a complete and realistic network for water treatment, storage, and distribution. The system operated continuously for 16 days, which included 14 days of normal operation and 2 days of attacks. During these attacks, the system was attacked 15 times.
- 3) *BATADAL* [36]: BATADAL dataset represents a WADI network consisting of seven storage tanks. The training set was generated from a year-long simulation and does not contain any attacks. The test dataset contains three months of continuous recordings and includes seven attacks.

In the experiments, only continuous time series are chosen during the dataset preprocessing stage. Then, 25, 62, and 27 variables are actually used in SWaT, WADI, and BATADAL, respectively.

B. Parameter Settings

STMBAD is implemented with Python 3.6.5. In our work, 20% of the training set is set aside to be used as the validation set, while the remaining 80% is used for model training. The training set is employed to initially train the model, and the validation set is used to evaluate the performance of different combinations of hyperparameters.

The hyperparameters of the model are carefully adjusted to ensure optimal performance for each dataset, as shown in Table II. The training set size of BATADAL is only 8761, which is too small for deep model training. Table III shows that in the optimization experiment of BATADAL data, the loss is the lowest when $Bs = 1$. This means that only one sample is used in each training iteration to achieve higher flexibility in model training with small datasets. The learning rate ε is adjusted using an exponential decay method as $\varepsilon = 0.5^{\text{epoch}-1} \varepsilon_i (i = 0, 1, \dots, 100)$

TABLE III
COMPARISON OF HYPERPARAMETERS OPTIMIZATION THE BATCH SIZE IN THE BATADAL DATASET

Batch sizes (Bs)	Metrics	Bs = 1	Bs = 16	Bs = 32
	Loss	0.0333	0.0728	0.1178
	Metrics	Bs = 64	Bs = 128	Bs = 256
	Loss	0.185	0.2721	0.3559

The bold to emphasize the data and to make them easier for the reader to read and analyze.

with an initial learning rate $\varepsilon_0 = 0.0001$, a decay coefficient of 0.5, and training epochs of 100 for each dataset.

In the anomaly detection phase, a sliding window is used to extract dynamic and localized information. To determine the optimal sliding window size and sliding step size corresponding to each dataset, several tests are performed. Fig. 6(a) shows the tuning of w with a fixed m , it can be seen from that the model performs best when $w = 205$ or $w = 210$. Then, m is adjusted when w is 205 or 210, respectively. Fig. 6(b) shows that the F1 score of the model obtains its maximum value when $w = 205$ and $m = 20$. Using the same methodology, the detection parameters of $w = 50$ and $m = 10$ are obtained under the SWAT dataset, and $w = 135$ and $m = 10$ are obtained under the WADI dataset.

This article implemented several baseline algorithms, including transformer, PatchTST, and TimesNet-based anomaly detection methods. The experimental results of these baseline algorithms are then compared with that of STMBAD.

- 1) *Transformer* (2017, [11]) is a well-known neural network model based on the attention mechanism. In sequential data processing, the attention mechanism allows the model to attend to different time steps or elements of the sequence, giving more weight to relevant parts and reducing the influence of less important ones.
- 2) *PatchTST* (2023, [37]) is a transformer-based long time-series prediction model. This model can effectively deal with the length variation and imbalance of the time series. It achieves this by using a patch-based approach where the time series is divided into smaller patches, and then transformer architecture is applied to these patches for prediction.
- 3) *TimesNet* (2023, [38]) is a time-series-based anomaly detection model. It combines CNN and LSTM networks to capture local and temporal dependencies in time-series data, effectively discovering abnormal behavior in time-series data.

After extensive parameter tuning of the methods on the same dataset, the combination with the best performance is selected as the final training parameters. Meanwhile, all methods are fully trained to ensure the authenticity and reliability of the comparison.

C. Overall Detection Performance

1) *ROC and AUC*: To visualize the effectiveness of STMBAD, the degree of coverage between predicted labels and true labels for our model and all baselines at the current time interval is shown in Fig. 7. As can be seen from the figure, our model is able to cover all anomalous regions in the sample, while the other baselines have missed detections. Meanwhile, receiver operating

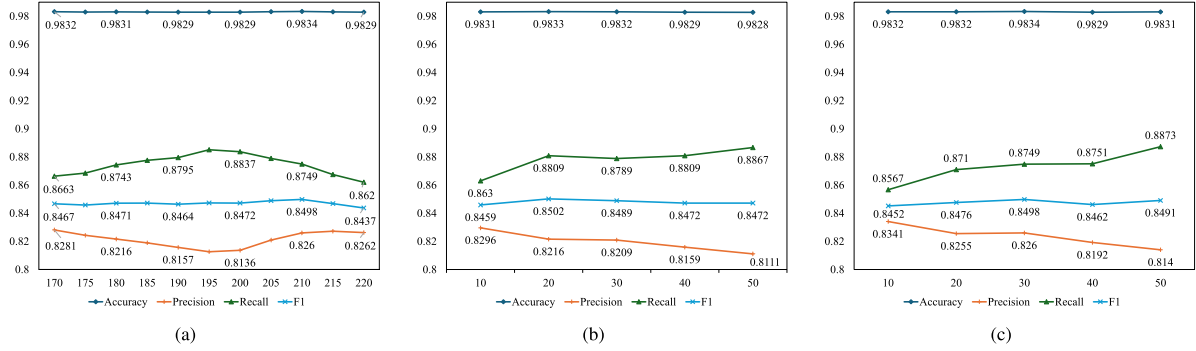


Fig. 6. Adjust the parameters of window size (w) and sliding step (m) size in the BATADAL dataset. (a) Window size ($m = 30$). (b) Sliding step ($w = 205$). (c) Sliding step ($w = 210$).

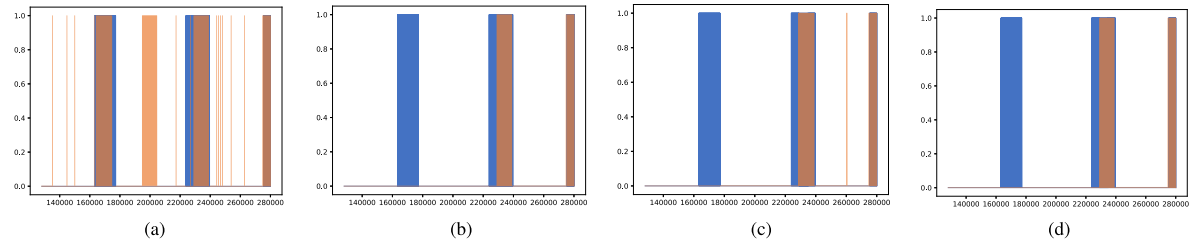


Fig. 7. Capability of anomaly detection on the BATADAL dataset. The blue region represents the real anomalous region, while the yellow color indicates the anomalous region detected by the corresponding model. (a) STMBAD. (b) PatchTST. (c) TimesNet. (d) Transformer.

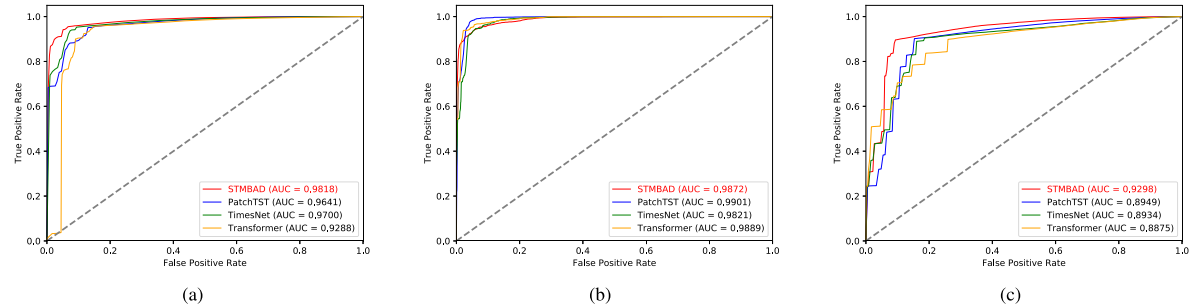


Fig. 8. Comparison of model performance. (a)–(c) ROC curves and corresponding AUC values under different datasets. (a) SWaT. (b) BATADAL. (c) WADI.

characteristic (ROC) curves are used to analyze the relationship between the false positive rate (FPR) and the true positive rate (TPR). The area under the ROC curves, namely AUC, is computed to evaluate the overall performance. As shown in Fig. 8, the AUC values of STMBAD on the BATADAL, SWaT, and WADI datasets are 0.8502, 0.9508, and 0.8717, respectively. The average AUC of our model is approximately 2% higher than that of the model in second place. In testing, there is a small bias in the optimal AUC. However, in anomaly detection, we prefer to obtain the best F1 value because having an excellent F1 value at a certain threshold is more important than having only high but less excellent F1 values at most thresholds [33]. The average values of FPR and TPR obtained by STMBAD in the optimal state are 0.92% and 90.69%, respectively. Most of the important metrics are higher than those of the baselines, which indicates that our classifier performs better than the baselines. The proposed method also achieves a high recall rate while ensuring a low false alarm rate.

2) Precision and F1-Score: The model proposed in this article is combined with several classical neural network frameworks for anomaly detection. Comparative experiments are conducted according to the original structure of the compared models or open source code. In Table IV, the F1 scores of our model on the SWaT, BATADAL, and WADI datasets are 0.9508, 0.8502, and 0.8717, respectively. The average precision rate of STMBAD is 0.8757, and the average recall rate is 0.9069, with the accuracy maintained above 98%. Compared with the average recall rate of the second-ranked model, the average recall rate has increased by 7.2%, and the average F1 has increased by 4.35%. This indicates that our model has high credibility and more comprehensive coverage in anomaly detection. Although the indicators of STMBAD on a single dataset are lower than those of the baseline model, it is still in an advantageous position from an overall analysis. In the SWaT dataset, the precision of transformer is about 4.69% higher than that of our model, but its recall rate is only 81.56%, which is 14.36% lower than

TABLE IV

COMPARISON OF ANOMALY DETECTION RESULT OF DIFFERENT METHODS UNDER THREE DATASETS

Datasets	Methods	Accuracy	Precision	Recall	F1
SWaT	Transformer	0.9766	0.9894	0.8156	0.8941
	TimesNet	0.9876	0.9398	0.9595	0.9496
	PatchTST	0.9841	0.9189	0.9534	0.9358
	STMBAD	0.9879	0.9425	0.9592	0.9508
BATADAL	Transformer	0.9623	0.7906	0.4062	0.5367
	TimesNet	0.9615	0.7356	0.4430	0.5530
	PatchTST	0.9710	0.7587	0.6735	0.7136
	STMBAD	0.9833	0.8216	0.8809	0.8502
WADI	Transformer	0.9814	0.9639	0.7046	0.8141
	TimesNet	0.9813	0.9314	0.7302	0.8186
	PatchTST	0.9878	0.9085	0.8778	0.8928
	STMBAD	0.9850	0.8630	0.8806	0.8717

The bold to emphasize the data and to make them easier for the reader to read and analyze.

TABLE V

COMPARISON OF DETECTION INFERENCE AND LATENCY TIME OF DIFFERENT MODELS UNDER THREE DATASETS

Datasets	Methods	Inference Time (s)	Delay Time (s)
SWaT	Transformer	6193	31.9359
	TimesNet	11529	3.4311
	PatchTST	6742	3.694
	STMBAD	3415	2.1836
BATADAL	Transformer	99	1.6456
	TimesNet	455	0.812
	PatchTST	96	0.9178
	STMBAD	63	0.7457
WADI	Transformer	1810	10.0551
	TimesNet	60256	8.9989
	PatchTST	18031	3.3154
	STMBAD	2542	1.3299

Inference time is the time between starting the test and getting the test result. Delay time refers to the length of deviation between the start time of a real anomaly and the start time of the most recently detected anomaly. The bold to emphasize the data and to make them easier for the reader to read and analyze.

that of STMBAD. And in the WADI dataset, the performance of our model ranks second. This is because the dataset has a higher frequency of sample collection, higher data density, and more serious data imbalance [39]. When training the model, the generalization ability of the model itself is insufficient, resulting in the test results being affected.

3) *Inference and Delay Time*: Across the three datasets under consideration, the inference time of STMBAD shows variation depending on the size of the dataset. As demonstrated in Table V, in the relatively small BATADAL dataset, the inference time is a mere 63 s. Moreover, on average, only 0.003 s are spent on inference per window. On the other hand, in the large WADI dataset, the inference time amounts to 2542 s. The average inference time over the three datasets is 2006 s, which is approximately 25% faster than that of the model ranked second. Evidently, STMBAD can perform detection rapidly and producing results in a timely fashion. It significantly enhances detection efficiency and saves valuable time costs. When compared from the perspective of latency time, it becomes clear that STMBAD is consistently at the fastest response level. It has an average

TABLE VI

COMPARISON OF F1 SCORES UNDER DIFFERENT METHODS

Method	SWaT	BATADAL	WADI
STMBAD(without adaptive threshold)	0.9034	0.8432	0.5980
STMBAD	0.9508	0.8502	0.8717

The bold to emphasize the data and to make them easier for the reader to read and analyze.

latency time of only 1.41 s. This model is 46.64% quicker than the one ranked second. This indicates that STMBAD is highly capable of detecting anomalies in a timelier fashion. It reduces the time difference between the occurrence of anomalies and their detection. In practical applications, this timely detection is crucial as it allows for prompt action to be taken to address the anomalies and prevent potential negative consequences.

4) *Fixed Versus Adaptive Threshold*: In this section, we will analyze the effectiveness of the proposed adaptive threshold setting method. The F1 scores of STMBAD on three different datasets are presented in Table VI. When the model is without the adaptive threshold, it obtains an average F1 score of approximately 0.7815. In contrast, our model with the adaptive threshold maintains an average F1 score at around 0.8909, representing an increment of 10.94%. This clearly demonstrates that the scene adaptive threshold determination algorithm is highly effective in identifying anomalies. Most existing algorithms typically only consider the global characteristics of the time series while neglecting the local trends of the data. As a result, this leads to an increase in the false alarm rate of some detection results. On the other hand, our model operates by computing local anomalies through sliding a window over the data series. In addition, it updates the global anomaly threshold based on the distribution of the data. By doing so, our model is able to capture the changing trends and periodicity of the data, thereby obtaining a more desirable detection result.

Upon analyzing the abovementioned four dimensions, it becomes evident that the performance of our model under different simulation datasets is at a high level. This proves that STMBAD has the ability to effectively address the problem of anomaly detection in multimodal spatio-temporal sequences.

V. CONCLUSION

This work propose an ICS anomaly detection approach, called STMBAD, based on spatio-temporal multimodal behavior. STMBAD leverages the rich information available in industrial multimodal data to deepen the understand of complex ICS behaviors and enhance the ability of detecting stealthy attacks. The proposed approach first embeds individual modality sequence separately into variate tokens, then uses the attention mechanism and FFN to capture multivariate correlations and interdependencies. Meanwhile, it utilizes the attention mechanisms to model the characteristics of the spatio-temporal multimodal behavior of ICS. STMBAD performs anomaly detection by evaluating the deviation of reconstructed variate tokens and leverages an adaptive detection mechanism combining global and local detection for variable industrial process. The simulation results show that the proposed method can effectively identify the anomalies in the ICSs and show high performance in anomaly detection tasks. STMBAD outperforms the baseline methods in many aspects, including average AUC value, average recall rate, average accuracy, and average F1 score. Moreover,

the average detection duration per window of STMBAD is 0.008 s, and the average latency is 1.41 s, which indicates that STMBAD responds much faster than the baseline systems.

In the experiment, it is found that the proposed model is more sensitive to the quality, size, and distribution of the data. For instance, when the data sample is unbalanced and the distribution undergoes drastic changes, the performance of the model will be affected. This indicates that the model's generalization ability is relatively degraded when facing with frequent changes in data distribution. These limitations are the mainly aspects being supposed to improve in future. By addressing these issues, the performance and applicability of the proposed model can be further enhanced in real-world anomaly detection scenarios.

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